

Using Topological Data Analysis Techniques on The Alternative Vote Referendum

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Chapter 1

Intro

The Alternate Vote (AV) is a voting system which is known for its ability to give smaller parties more of a chance by combatting the spoiler effect which, in First Past the Post Elections, encourages voters to vote only for the parties with the best chance at winning. The AV and the similar Instant-Runoff Vote (IRV), are continuing to be adopted increasingly around the world such as in 2022 when Alaska started using the IRV in its General Elections.

In 2011 the UK had a referendum on the Parliamentary voting system to decide if General Elections would be conducted using the AV. The referendum was a landslide no, and the UK, still uses the generally considered inferior First Past the Post system.

In this paper we will analyse the factors that impacted the outcome of the referendum in the hope that it can uncover factors that could affect the referendum if it happened again. Indeed, The 2025 British Social Attitudes survey [4] found that support for electoral reform has risen to 60% suggesting that another referendum is possible and possibly winnable.

We will use the Topological Data Analysis technique called the Ball Mapper algorithm similarly to [5] to capture relationships in the data as it can effectively capture non-linear relationships.

Chapter 2

Summary of "An Economic Topology of the Brexit Vote"[5]

The paper acknowledges that "there are many factors combining to produce the observed Leave vote ... [which] pose important statistical challenges that must be overcome. To this end analyses premised on linear relationships, or bivariate considerations, are prone to missing important elements of the story. The paper opts to use TDA to analyse Brexit and its factors, as "it is free from assumptions about relationships and can ... capture coordinates on any number of axes to fine grain all possible interactions". Indeed, TDA has an important robustness that distributional methods such as regression do not". Moreover, "Against the discourse in the literature such an ability is vital to facilitate a strong evidence base".

The paper takes "Parliamentary constituency data ... from the British Election Study" combined with "an estimate of the percentage of voters who selected leave and remain in each constituency" to understand the "economic make-up of each region". Which specifically provides a link between parliamentary voting habits and those witnessed in Brexit albeit based on "the assumption that the demographic make up of each constituency did not vary greatly" between the General Election and the Brexit Referendum, a "commonplace assumption in the literature". The paper analyses the data using statistical methods such as quartile-based comparative analysis and t-tests to study a subset of parameters at a time such as: The 2015 General Election, Tenure, Household Occupants, Qualifications and Deprivation. This serves as a basis for the further analysis throughout the paper as they provide indicators of what is to be expected for example "those with degrees are found far more in Remain constituencies, comprising a third of all households there in and so, a good model should acknowledge this increase in remain votes with people who have degrees.

Since ball mapper has been previously explored we shall we shall focus our efforts on how the paper made use of the algorithm. The first thing that the paper considers is

selecting a suitable ball radius ϵ . The paper begins by varying some different values for ϵ . As some sampling data to get a good idea of a sufficient ϵ , the paper just takes the vote share three largest political parties and then a separate axis that considers the vote share of all of the other parties. A couple observations were highlighted, as ϵ increases the number of balls decreases (as expected), when $\epsilon = 5$ we find there are many balls that are disconnected containing only one constituency and on the other end when $\epsilon = 20$ there are 17 balls and no more disconnected constituencies. Another observation as reported by 2.1 is that as ϵ increases the number of edges per ball tends towards 2. The paper settled on $\epsilon = 12$ for "a good balance of detail and readability".

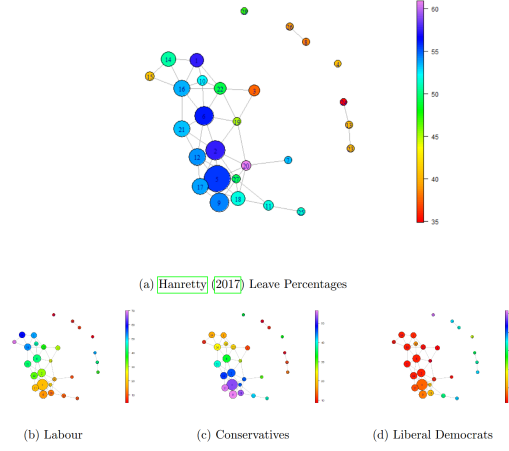
Table 2: Radius and Ball Size: 2015 Vote Percentages

ϵ	Balls	Size (Mean)	Size (sd)	Edges	ϵ	Balls	Size (Mean)	Size (sd)	Edges
1	585	1.084	0.301	0.005	15	25	51.08	52.23	1.720
2	420	1.709	1.238	0.190	20	17	78.29	73.56	1.882
5	165	6.412	7.618	1.067	25	11	113.8	114.9	1.818
7	98	12.29	15.21	1.602	30	8	144.1	138.6	2.125
10	53	23.25	26.30	2.057	35	7	185.3	154.5	2.143
12	35	33.46	39.47	1.686	40	5	240	137.0	1.800

Figure 2.1: A table demonstrating the different characteristics the ball mapper graph has when varying different values of epsilon.

The paper goes on to analyse the the ball mapper for the 2015 voting data. The paper starts by looking at the overall shape and notices that it has one main shape that has four axis and three separated segments. The paper makes the observation that the middle section of the main shape has the majority of the leave votes and the top two axis and the separated segments are higher in remain votes. Using 2017 voting data for each of the parties the paper highlights that the top of the centre of the main shape is mainly Labour affiliated and the bottom of the centre of the main shape being majority Conservative affiliated. Also, the right two axis are where the Liberal Democrats control the votes and the left two axis as well as the separated segments are where the Other parties (such as the Scottish National Party and the Plaid Cymru) control the votes. The paper highlights some interesting interaction between the balls, such as ball 16 which has strong conservative support but low leave votes. Other key observation seem to be that in the more Liberal Democrat balls such as 17, 29 and 30 have "much lower" leave percentages.

Figure 5: 2015 Voting: Three Parties ($\epsilon = 12$)



Section 5 of the paper applies much of the method of analysis as the party data, however, the difference is that it focuses on other data that may have affected leave votes. The data that was used was 2011 census variables consisting of tenure, household composition, number of motor vehicles, NSSEC Classification, Qualifications, self reported health and deprivation level. Each variable uses a different value of ϵ in order to make the data being studied clearer.

Figure 14: Full Dataset: $\epsilon = 18$

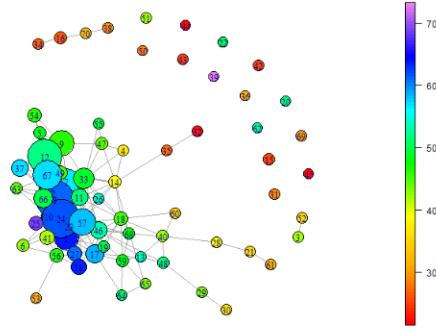


Figure 2.2: Ball mapper graph showing the full dataset

Figure 1 (Figure 14 from [5]) shows a bloc of large blue connected balls in the centre of a connected shape, lighter blue balls and green ones are observed to surround the blue bloc. Spiking out of the shape while still being connected are some yellow and orange balls representing important marginal constituencies. Finally, there is a significant amount of

mostly red remain unconnected balls or with very little connections and not to the main shape.

The report looks at 3 main comparisons from this graph. First, they looked at the string of strong remain balls connected to the main shape. The red and orange balls connect to the shape via a yellow one. The remain balls had lower outright ownership and higher number of private renters in comparison to the yellow (ball 14).

The second comparison is between balls 53 and 56 (bottom left of connected component, orange and green). There are not too many differences seen here however, in ball 53 contingencies there are more living alone and it has a higher level 4 deprivation level than ball 56. This is contrary to the idea that all poor areas voted leave. It may be an outlier though as ball 53 only has 2 constituencies (ball 56 has 16).

The final comparison is between balls 14 and 26 (yellow, blue-green). Ball 26 has a 55% leave vote, 15% higher than ball 14. There are 8 characteristic differences which the 9 constituencies of ball 14 have with the 5 constituencies of ball 26. Ball 14 has more students and less pensioners than 26. Ball 14 also has more a levels (level 3), while 26 has more holders of the semi-routine NSSEC status and less healthy constituents. All these factors lead to the accurate suspected conclusion that ball 14 voted to leave the EU and ball 26 voted to stay. Note that there are a large amount of characteristics for which the balls compared have similar results. Other things could be looked at such as why there are so many connected arms. Regional variations and the role of voting patterns.

TDA ball mapper is very good at linking collective characteristics with an outcome of an interest. However, it is normally quite hard to make an ordinary least squares model from it, yet with the reduced dataset it is now possible to compare it to TDA insights.

As seen in other studies, region tends to effect the Brexit vote, specifically London vs “Not London” and that the east midlands and east of England voted to leave more than expected and may have been the deciding votes. Plotting the ball mapper graphs for each region using the reduced dataset give us 12 graphs. London and Scotland are seen to be the strong remain voters, their graphs are similar showing a large group of large red balls with a mix of colours spiking out. The east of England’s largest ball had its highest leave vote proportion, as does the east midlands and Wales which counters the industrial north of England narrative for the Brexit result.

Overall a lot of the narratives behind Brexit are far from proven, the left behind industrial north claim is simplistic and lacks to take account of non-linearities in the data. This paper found that remain areas tend to be very similar while leave areas are diverse. TDA ball mapper graphs show that the data is very complex, and the current work is missing a lot of the picture.

Chapter 3

TDA

In this section we analyse data regarding the 2011 Alternate Vote Referendum. We are going to first analyse the ball mapper of this data and then proceed to analyse this data along with census data taken from the time.

3.1 Data

Before looking into the 2011 alternative vote referendum in the UK we investigated the 1998 good Friday referendum in the republic of Ireland and Northern Ireland as well as the devolution referendum in Wales in 1997. We encountered problems in making a ball mapper analysis of both. For the good Friday referendum, constituency based data only exists for the republic as all votes were counted centrally in the Northern Ireland. For the Welsh referendum we successfully collected data on political allegiance, Welsh identity and population density. This included converting the 1997s elections constituencies into the Welsh councils. We completed ball mapper analyses but found the data didn't have enough points for meaningful graphs as Wales has only 22 regions.

The final and successful approach was to follow the paper's example [5] by using a whole-UK based referendum, specifically the alternate vote referendum. Data is collected from three sources: the alternate vote data taken from an archive of the electoral commissions website [1], the 2010 General Election results from the House of Commons Library [2], and, the 2011 census data from the Office For National Statistics [6].

Unfortunately the General Election and the Alternate Vote Referendum are grouped by different voting areas. This makes combining the data tricky as the AV voting areas can't be given concrete political data. Because of this we have chosen to keep only those voting areas that were precisely the same between the AV and GE, as the accuracy of their data will provide the best results. Unfortunately this means losing around 60% of our potential data points, but there are still enough for ball-mapper to be insightful. We also checked that the consistent voting areas had no obvious trends that could impact

our analysis, such as the majority being from the same geographical area.

Four python scripts were written and are included in AppendixA.2 three of which format the individual tables, removing unnecessary columns and formatting the kept columns, and the fourth which compiles all the tables into one.

3.2 Political Parties

The most important step is choosing an appropriate ϵ for the ball mapper graph, as a poor choice of ϵ can miss the broad trends in the data, or the more localised trends. Having reviewed multiple graphs, $\epsilon = 0.33$ has been chosen as it best illuminates the trends in the data.

Figure 3.1 shows not a single ball has percentage yes of more than 50%, with most being around 30%, the vote was a landside, unlike the brexit vote [5]. The balls with higher yes vote percentage are located near the top of the main shape and in the disconnected outliers of the graph, balls 15, 13, 9 and 3, except for disconnected ball 12, which has the lowest vote percentage on the graph.

Alongside this graph, consider figures 3.2 which are coloured by the percentage of people who voted for a particular party in each ball. Examining the main shape, we see areas where certain categories live, the bottom is generally conservative, the middle to the top is mainly Labour, whilst the Liberal Democrats are generally at the bottom left and in disconnected balls 13 and 9. The other parties have a strong presence in the disconnected balls, and a minor presence at the top of the shape. A notable ball is ball 12 which is a disconnected and mostly conservative, the only conservative ball that is disconnected from the shape.

A strong correlation can be seen between where other parties are, and where the vote was more pro the AV than in other areas. Indeed, both can be found at the top of the main shape, and in the outliers. Furthermore there is a correlation of where the conservatives are with where the vote was more strongly against. These are unsurprising as they align with the existing literature[3]. The conservative party took a stance against the alternate vote, hence their likelihood to strongly disagree. Since the AV could potentially benefit smaller parties by weakening the dominance of the major parties, the other parties which include all the smaller parties ought to be generally pro the AV.

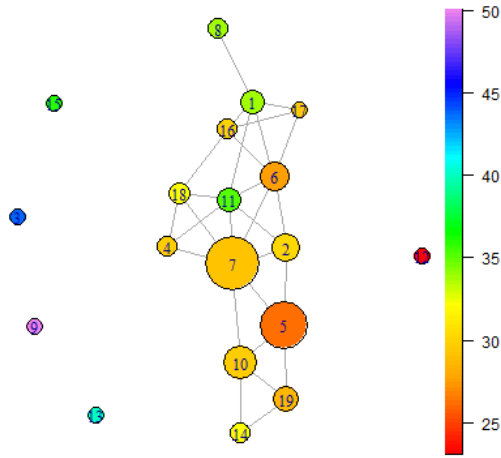


Figure 3.1: A ball mapper graph coloured by the percentage yes vote and drawn based on the percentage party vote alongside the percentage yes vote for $\epsilon = 0.33$.

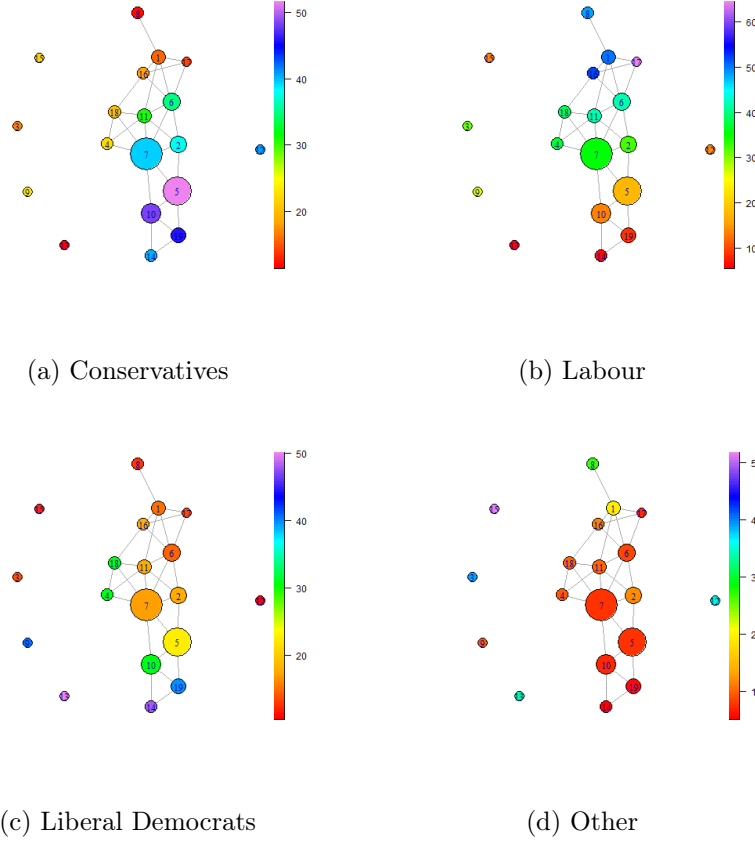


Figure 3.2: A figure that contains four ball mapper graphs coloured by percentage of occupants who voted for each political party in the 2010 General Election. The graph was created using the same data and $\epsilon = 0.33$ as 3.1.

3.3 The Full Dataset

We aim to discover if debates on electoral reform are partisan, socio-economic or generational in nature. Do demographic variables have an impact on the 2011 UK Alternative Vote referendum? Our full dataset looked at three different possible factors that influenced the 2011 alternative vote referendum result. Political allegiance based of the 2010 general election shown above, as well as levels of education and age demographics. We suspect the more educated and younger a voter is the more likely they are to vote for the proposed reforms. Also seen above was our suspicion that Conservative voters voted for against the reform, while smaller party voters voted for it.

Ball mapper's multidimensionality is useful here, we can easily see the relationships between variables and may uncover unexpected clusters. We used R to analyse the data columns A.3 and produce the following graph.

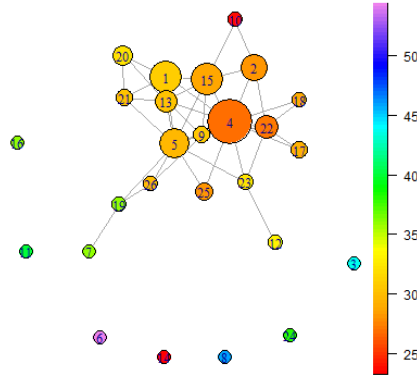


Figure 3.3: Ball Mapper Graph made using all the columns A.3 coloured by the percentage of votes for yes in the AV referendum, $\epsilon = 0.74$

An epsilon of 0.74 was used and the colours indicate what percentage of a ball (group of constituencies) voted for AV. There is a large cluster of mostly orange balls in the upper centre right. In and around 30% voted yes in these balls. There are also some very large balls, in particular ball 4. This is a very density populated area in the data space and contains a large amount of conservative and labour voters. Other parties had a very limited presence in ball 4. The average age is 40.

There are a notable number of unconnected balls scattered below the main cluster ranging in colour. 3 Green, 2 Blue, 1 pink and 1 red. The red stands out as the only safe No ball that is isolated. Ball 14 also has one of the strongest no votes. It seems to be the conservative heartland with a very low percentage of voters having allegiance to any other party. Additionally, ball 14 is one of the oldest with a very large over 60 population. The mean age, around 50. is the highest we recorded. Ball 6 on the other hand is the only pink ball in the whole graph, which makes it the only ball to vote yes. We found that it has very high rates of level 3 and 4 qualifications, that is A levels and above. Also, in comparison to ball 14, 6 has one of the youngest populations with a mean age of just over 35.

A few spikes can be observed coming out of the main cluster, balls 23 and 12 as well as ball 10 are examples. However we will look at balls 26, 19 and 7 which go from orange to green indicating more subtle differences in demographics than in ball 14 and 6. All three balls have a similar mean and median age of just above 35, however 7 looks to be closer to 40. Ball 19 has mediocre rate of level 1 and 2 qualifications, ball 26 a slightly higher amount while ball 7 has a low amount. Level 3 is the opposite with ball 7 having slightly more than 19 and 26 has a lower amount than both. Notably, ball 7 has the

highest rate of level 4 while 26 has a medium amount and 19 not too far below that. And while all have a roughly equal proportion of no and other qualifications, apprenticeships were highest in 26 and lowest in 7. Ball 10 may represent part of Wales or Scotland due to the large presence of other parties category. We would expect them to vote yes as smaller parties would benefit from AV, however they voted very strongly for no. This implies it is parties like UKIP people voted. for but if this is true it is surprising ball 10 has similar demographics to the main cluster.

As done in the Brexit report, we looked at t-tests to see what variables we expect to determine the result. These were also done in R. Now there was an insufficient amount of yes votes to do t-tests on the referendum results so instead we examined whether variables could predict if a constituency will be a “no” region (less than 40% yes) or a “lean no” region (40% to below 50%).

Variable	Mean LeanNo	Mean No	Diff	SD	Min	Max	p-value	Sig
Votes	34395.67	46111.21	-11715.55	7450.98	26078.0	70264.0	0.1134	
Cons	8119.33	18352.56	-10233.23	7915.67	1993.0	32810.0	0.0353	*
Lab	11740.67	12622.96	-882.29	5747.69	2210.0	31650.0	0.7745	
LD	8609.33	10616.46	-2007.13	5589.60	3285.0	28531.0	0.6103	
Other	63076.67	76350.92	-13274.26	8742.70	52715.0	119267.0	0.1141	
PercentageYes	43.53	29.26	14.27	4.69	20.3	45.7	0.0052	**
No qualifications	20.00	23.83	-3.83	5.14	12.7	37.8	0.3686	
Level 1	10.30	14.06	-3.76	1.82	8.1	18.4	0.0857	
Level 2	13.10	16.15	-3.05	1.17	11.0	18.4	0.1053	
Apprenticeship	2.80	4.02	-1.22	0.76	1.9	5.8	0.1144	
Level 3	17.87	12.24	5.62	2.11	9.6	25.9	0.3545	
Level 4+	31.00	24.92	6.08	6.37	14.0	44.5	0.0535	
Other	4.97	4.79	0.18	1.28	3.3	13.7	0.7397	
Age: Mean	36.53	40.51	-3.98	2.38	33.5	48.9	0.1179	
Age: Median	33.67	40.82	-7.15	3.62	27.0	52.0	0.1642	

Figure 3.4: Graph showing T-test of each variables base on "No" and "LeanNo"

From the table we can deduce that the only significant variable at the 5% level in predicting if a region will be a safe no spot or if it will only lean no is the conservative vote. Areas with a higher conservative vote, were more likely to be strong no regions which aligns with what we found in the party analysis. The lack of variables strongly indicating no vs lean may explain why there are so many edges within the central orange cluster in the ball mapper. With a moderate average age and the typical mix of education levels, a broad cluster with no clear splits in the core of the no vote is expected. We did a second set of t-tests where we changed Lean no to be anything over 30%. In this case, the conservative and other party vote were extremely significant (0.001%), as well as the obvious percentage yes and level 1,2 and apprenticeship levels of education. Median age was significant at a 0.1% level while mean age and level 4+ qualification levels were significant at 5%. While we note these results, their usefulness is limited as they are all significant in predicting if a area which voted against the reform has below 30% of its population voting for it. There is a large amount of sensitivity in the results of these tests depending on how we define lean no.

Chapter 4

Conclusion

We have applied many of the techniques used in the analysis of the Brexit Referendum [5] to the Alternate Vote referendum and, using the Ball Mapper algorithm's ability to effectively capture non-linear relationships between many variables, have analysed the correlation between General Election voting habits, Qualifications, and ages to find notable correlations.

We have deduced some of the factors that are most likely to influence a persons vote, the largest being the Conservative tendency to vote strongly against followed by the other parties' tendency to vote weakly against. Which we have confirmed through a dedicated party analysis, the full ball-mapper analysis, and T-tests.

Overall we have demonstrated the efficacy of the Ball Mapper algorithm when applied to political data, in particular furthering its use for referendum analysis as [5] have done before.

Bibliography

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- [5] Rudkin Dlotko and Qiu. “An Economic Topology of the Brexit Vote”. In: (2019).
- [6] Office for National Statistics. *2011 Census Data Catalogue*. Catalogue listing 2011 Census tables by type, topic, and release, including population and household estimates and detailed characteristics. Office for National Statistics. 2025. URL: <https://www.ons.gov.uk/census/2011census/2011censusdata/2011censusdatacatalogue> (visited on 12/12/2025).

Appendix A

Appendix

A.1 R Code for Ball-Mapper Algorithm

```
library("BallMapper")
library("magick")

# Read in data
setwd(".")
pts <- read.csv("./Data/all.csv")

# Make names valid identifiers (replaces spaces with dots)
names(pts) <- make.names(names(pts))

# PARTY BALL MAPPER ANALYSIS


---


# Extract the parties
party_names <- c("Percentage.Yes", "Percentage.Con", "Percentage.Lab", "
  Percentage.LD", "Percentage.Other")

# Create image directories
dir.create("Party-Analysis")
lapply(paste0("Party-Analysis/", party_names), dir.create)

# Extract and Normalize party data
party_data <- pts[party_names]
party_data <- normalize_to_min_0_max_1(party_data)

# Create graphs for each party's colouring and a variety of epsilon
for (e in 10:80) {
  for (p in party_names) {
    epsilon <- e / 100
    coloring <- pts[p]
    # Set the seed so diagrams look the same
    set.seed(12345)
    # Create the plot in the correct folder
```

```

    graph <- BallMapper(party_data, coloring, epsilon)
    ColorIgraphPlot(graph, store_in_file = paste0("Party-Analysis/", p, "/"
    , e, ".jpg"))
    # coloredDynamicNetwork(graph)
  }
}

# Create gifs out of the graphs
dir.create("Party-Analysis/Gifs")
for (p in party_names) {
  m <- image_read(paste0("Party-Analysis/", p, "/", 10:80, ".jpg"))
  m <- image_animate(m)
  m <- image_write(m, paste0("Party-Analysis/Gifs/", p, ".gif"))
}

# Function which plots a party for a specific value of epsilon
plot_for_epsilon <- function(e, p) {
  # Create Epsilon Directory
  dir_name <- paste0("./Parites-", e)
  dir.create(dir_name, showWarnings = FALSE)
  epsilon <- e / 100
  coloring <- pts[p]
  set.seed(12345)
  graph <- BallMapper(party_data, coloring, epsilon)
  ColorIgraphPlot(graph, store_in_file = paste0(dir_name, "/", p, "-", e, "
  .jpg"))
  return(graph)
  # coloredDynamicNetwork(graph)
}

# Use plot_for_epsilon to create specific graphs
# Plot a specific graph
# graph <- plot_for_epsilon(63, "Conservative")
# Plot all parties for one epsilon
# graphs <- lapply(party_names, plot_for_epsilon, e = 28)

# FULL BALL MAPPER ANALYSIS -----
# Extract all numeric columns and normalize
numeric_cols <- sapply(pts, is.numeric)
full_pts_norm <- normalize_to_min_0_max_1(pts[, numeric_cols])

# Choose epsilon
e <- 74
epsilon <- e / 100
# Create folder for all the images
dir.create(paste0("./Full-", e), showWarnings = FALSE)

# Generate coloured images
for (col in colnames(full_pts_norm)) {
  coloring <- pts[col, drop = FALSE]
  # Fix seed so drawing is the same
  set.seed(12345)

```

```

graph <- BallMapper(as.data.frame(full_pts_norm), coloring, epsilon)
# Create a safe name from the column title
safe_name <- gsub("[^[:alnum:]_.-]", "_", col)
ColorIgraphPlot(graph, store_in_file = paste0("./Full-", e, "/",
  safe_name, ".png"))
}

```

A.2 Python Code for Combining Tables

A.2.1 Code for Age data

```

import pandas as pd

# Read raw age data
df = pd.read_csv("age-raw.csv")

# Drop columns
df = df.drop(["date", "geography_code", "Rural_Urban"], axis=1)

# Rename age columns
renameMap = {}
for c in df:
    if "Age" in c:
        newName = c.split(":")[1].split(";")[0].strip()
        newName = newName.replace("_", ".")
        renameMap[c] = newName
df = df.rename(columns=renameMap)

toPercentage = df[['Age.0.to.4', 'Age.5.to.7', 'Age.8.to.9', 'Age.10.to.14',
    'Age.15', 'Age.16.to.17', 'Age.18.to.19', 'Age.20.to.24', 'Age.25.to.29',
    'Age.30.to.44', 'Age.45.to.59', 'Age.60.to.64', 'Age.65.to.74', 'Age.75.to.84',
    'Age.85.to.89', 'Age.90.and.over']]
x = []
for col in toPercentage:
    x.append("Percentage." + col)
    df["Percentage." + col] = (df[col] / df["All.usual.residents"]) * 100
df = df.drop(toPercentage, axis=1)

# Drop unnecessary columns such as under 18s
df = df.drop(["All.usual.residents", 'Percentage.Age.0.to.4', 'Percentage.
    Age.5.to.7', 'Percentage.Age.8.to.9', 'Percentage.Age.10.to.14', '
    Percentage.Age.15', 'Percentage.Age.16.to.17'], axis=1)

df.to_csv("./age-formatted.csv", index=False)

```

A.2.2 Code for Alternate Vote data

```

import pandas as pd

# Read data

```

```

df = pd.read_csv("./Alternate-Vote.csv")
df = df[["Areas", "%_Yes", "%_No", "Turnout"]]

# Format the percentages from '67%' to '67'
df["Percentage.Yes"] = (df["%_Yes"]).str.replace("%", "", regex=False)
df["Percentage.No"] = (df["%_No"]).str.replace("%", "", regex=False)
df["Percentage.Turnout"] = (df["Turnout"]).str.replace("%", "", regex=False)
df = df.drop(["%_Yes", "%_No", "Turnout"], axis=1)

df = df.drop(["Percentage.No", "Percentage.Turnout"], axis=1)

df.to_csv("./AV-Formatted.csv", index=False)

```

A.2.3 Code for General Election 2010 data

```

import pandas as pd

df = pd.read_csv("./GE2010-results.csv")

# Remove the Northern Irish rows:
oldLen = len(df)
df = (df[df["Region"] != "Northern_Ireland"])
print(f"Removed: {oldLen - len(df)} constituencies")

# Remove unwanted columns
df = df.drop(["Press_Association_Reference", "Election_Year", "Electorate"], axis=1)

keepColumns = ["Constituency_Name", "Region", "Votes", "Con", "Lab", "LD"]

# Frame to sum for other votes
sf = df.drop(keepColumns, axis=1)

# Remove unwanted columns
df = df[keepColumns]

# Calculate other votes
df["Other"] = sf.sum(axis=1)

# Calculate percentage votes for each party
party_cols = ["Con", "Lab", "LD", "Other"]
for col in party_cols:
    df[col] = (df[col] / df["Votes"]) * 100

# Rename columns to indicate percentages
df = df.rename(columns={"Con": "Percentage.Con", "Lab": "Percentage.Lab", "LD": "Percentage.LD", "Other": "Percentage.Other"})

# Drop now unneeded columns
df = df.drop(["Region", "Votes"], axis=1)

```

```
df.to_csv("./GE2010-Formatted.csv", index=False)
```

A.2.4 Code for merging all other data files

```
import pandas as pd

# Read in data
ge = pd.read_csv("GE2010-Formatted.csv")
av = pd.read_csv("AV-Formatted.csv")
qual = pd.read_csv("Qualifications-raw.csv")
age = pd.read_csv("age-formatted.csv")
l = len(ge)

# Merge first 2 tables
df = pd.merge(
    ge,
    av,
    left_on="Constituency_Name",
    right_on="Areas",
    how="inner",
    indicator=True
)

# Record the number of elements lost from the left table
print(f"{l-len(df)}_mismatches")

# Add Qualifications
df = pd.merge(
    df,
    qual,
    left_on="Constituency_Name",
    right_on="parliamentary_constituency_2010"
)

# Record the cumulative number of elements lost from the left table
print(f"{l-len(df)}_mismatches")

# Add ages
df = pd.merge(
    df,
    age,
    left_on="Constituency_Name",
    right_on="geography"
)

# Record the cumulative number of elements lost from the left table
print(f"{l-len(df)}_mismatches")

# Remove merged columns
df = df.drop(labels=["Areas", "_merge", "parliamentary_constituency_2010",
                    "All_categories"], axis=1)
```

```
df.to_csv("./all.csv", index=False)
```

A.3 Data Columns Used

The columns of data that were used in the ball mapper analysis are:

- Constituency Name
- Percentage Con
- Percentage Lab
- Percentage.LD
- Percentage Other
- Percentage Yes
- No qualifications
- Level 1 qualifications
- Level 2 qualifications
- Apprenticeship
- Level 3 qualifications
- Level 4 qualifications and above
- Other qualifications
- School children and full-time students: Age 18 and over
- geography
- Mean Age
- Median Age
- Percentage Age 18 to 19
- Percentage Age 20 to 24
- Percentage Age 25 to 29
- Percentage Age 30 to 44
- Percentage Age 45 to 59
- Percentage Age 60 to 64
- Percentage Age 65 to 74
- Percentage Age 75 to 84

- Percentage Age 85 to 89
- Percentage Age 90 and over